



CHEMISTRY HSSC-I

SECTION – A (Marks 17)

Time allowed: 25 Minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent.

Deleting/overwriting is not allowed.

Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی طریق پر دے کر نام مرکز کے حوالے کریں۔ گالت کر دو بار
لکھنے کی اجازت نہیں ہے۔ سیاہ پینسل کا استعمال ممنوع ہے۔

Version No.				
3	0	0	5	1

ROLL NUMBER						

0	0	0	0	0
1	1	1	1	1
2	2	2	2	2
3	3	3	3	3
4	4	4	4	4
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8	8	8	8	8
9	9	9	9	9

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1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9

Answer Sheet No. _____

Invigilator Sign. _____

Fill the relevant bubble against each question according to curriculum:

Candidate Sign. _____

Question	A	B	C	D	A	B	C	D
1. The third ionization potential value of an element is significantly high as compared to its 1 st and 2 nd ionization potentials. The element belongs to:	Group-1 (I-A)	Group-2 (II-A)	Group-17 (VII-A)	Group-18 (VIII-A)	☐	☐	☐	☐
2. Which pair of elements will most likely form an ionic bond?	C & H	B & Cl	Na & O	H & Cl	☐	☐	☐	☐
3. Which statement is FALSE about comparative bond energy values of Hydrogen halide molecules?	H-Cl > H-F	H-F > H-Cl	H-Cl > H-Br	H-Br > H-I	☐	☐	☐	☐
4. The number of molecules present in 67.2 dm ³ of He gas at STP is:	6.02 × 10 ²⁴	1.204 × 10 ²⁴	1.806 × 10 ²⁴	2.4 × 10 ²⁴	☐	☐	☐	☐
5. What will be the enthalpy of solution of NaCl if ΔH (reverse lattice) is +786 kJ/mol and ΔH (hyd) is -783 kJ/mol?	+3 kJ/mol	-3 kJ/mol	+1569 kJ/mol	-1569 kJ/mol	☐	☐	☐	☐
6. The following equilibrium will NOT be shifted in the forward direction by: (N ₂ + 3H ₂ ⇌ 2NH ₃ + Δ)	Increasing temperature	Increasing pressure	Adding H ₂ gas	Removing NH ₃	☐	☐	☐	☐
7. Aqueous solution of which of the following salts will have pH less than 7?	Na ₂ SO ₄	KNO ₃	CuSO ₄	Na ₂ CO ₃	☐	☐	☐	☐
8. Position of an element in periodic table with electronic configuration 1s ² , 2s ² p ⁶ , 3s ² p ² is:	In Group-1 (I-A)	In Group-2 (II-A)	In Group-13 (III-A)	In Group-14 (IV-A)	☐	☐	☐	☐
9. Peroxyacetyl nitrate (PAN) is formed by the reaction of oxides of nitrogen with:	Oxides of sulphur	Hydrogen sulphide	Oxides of phosphorus	Unburned hydrocarbons	☐	☐	☐	☐
10. Which of the following halides can reduce H ₂ SO ₄ to H ₂ S?	NaF	NaCl	NaBr	NaI	☐	☐	☐	☐
11. Identify the step in the mechanism of free radical chlorination of ethane: CH ₃ - CH ₃ + Cl [•] → CH ₃ - CH ₂ [•] + HCl	Initiation	Propagation	Substitution	Termination	☐	☐	☐	☐
12. Identify the product formed when Ethyl Bromide is warmed with NaOH in ethanol:	CH ₂ =CH ₂	CH ₃ -CH ₂ -ONa	CH ₃ -CHO	CH ₂ =CH-Br	☐	☐	☐	☐
13. Which reagent is used for the reduction of Carboxylic acids to Alcohols?	KMnO ₄	K ₂ Cr ₂ O ₇	LiAlH ₄	H ₂ /Pt	☐	☐	☐	☐
14. The compound which gives Silver mirror test as well as Iodoform test is:	Ethanol	Ethanal	Propanal	Propanone	☐	☐	☐	☐
15. Which pair of organic compounds can be distinguished by using 2, 4-DNPH?	Ethanol and Acetone	1-Propanol and 2-Propanol	Ethanal and Ethanol	Ethyl chloride and Ethyl bromide	☐	☐	☐	☐
16. Identify correct relationship for determination of half-life of a first order reaction:	t _{1/2} = 0.693 / k	t _{1/2} = [A ⁰] / 2k	t _{1/2} = 1 / k[A ⁰]	t _{1/2} = 0.023 [A ⁰] / k	☐	☐	☐	☐
17. Which of the following organic compound can be identified by Fehling solution test?	CH ₃ -CH ₂ -OH	CH ₃ -COOH	CH ₃ CH ₂ CHO	CH ₃ -CH ₂ -Cl	☐	☐	☐	☐

—1HA-I 25005(B)—

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SUPPLEMENTARY TABLE

Atomic No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Symbol	H	He	Li	Be	B	C	N	O	F	Ne	Na	Mg	Al	Si	P	S	Cl	Ar	K	Ca
Mass No	1	4	7	9	11	12	14	16	19	20	23	24	27	28	31	32	35.5	40	39	40



Total Marks Sections B and C: 68

(14 x 3 = 42)

SECTION – C (Marks 26)

Q.3	What is mass spectrometry? Explain the construction and working of a mass spectrometer. How can average atomic mass of an element be calculated from data obtained?	1+4 +2	OR	Explain formation of covalent bonds in O_2 molecule on the basis of molecular orbital theory and valance bond theory. Justify that molecular orbital theory is superior to valence bond theory.	3+3 +1									
Q.4	Describe by giving balanced chemical equations and necessary conditions that how a Carboxylic acid can be prepared from: a) CH_3-CH_2-Cl b) $CH_3-CH-CH_2-OH$ CH_3 c) C_6H_5-CHO	2x3	OR	Discuss the effect of following changes on the given reactions at equilibrium:	2x3									
				<table border="1"> <tr> <td>i</td><td>Increasing pressure</td><td>$C_3H_8 + 5O_2 \rightleftharpoons 3CO_2 + 4H_2O$</td></tr> <tr> <td>ii</td><td>Removing SO_3</td><td>$2SO_2 + O_2 \rightleftharpoons 2SO_3$</td></tr> <tr> <td>iii</td><td>Decreasing temperature</td><td>$CO_2 + \Delta \rightleftharpoons CO + \frac{1}{2} O_2$</td></tr> </table>	i	Increasing pressure	$C_3H_8 + 5O_2 \rightleftharpoons 3CO_2 + 4H_2O$	ii	Removing SO_3	$2SO_2 + O_2 \rightleftharpoons 2SO_3$	iii	Decreasing temperature	$CO_2 + \Delta \rightleftharpoons CO + \frac{1}{2} O_2$	
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iii	Decreasing temperature	$CO_2 + \Delta \rightleftharpoons CO + \frac{1}{2} O_2$												
Q.5	How can one identify the functional groups in the following compounds by chemical tests? Elaborate. a) CH_3CHO b) CH_3-CH_2-OH c) $CH_3-CH=CH_2$	2x3	OR	Why is Ethene more reactive than Ethane? Write down the reactions of Ethene with: a) H_2SO_4/H_2O b) Cl_2/H_2O	2+2 +2									
Q.6	What is solubility product? Derive an expression for the solubility product of CaF_2 . How can the concept of solubility product be applied to predict the precipitation of CaF_2 on mixing aqueous solutions of Ca^{+2} and F^{-1} ions?	1+2 +4	OR	Explain the occurrence of a chemical reaction with reference to molecular collisions, activation energy, formation of activated complex and enthalpy changes in the reaction.	07									

— 1HA-I 25005 (B) —

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Symbol	H	He	Li	Be	B	C	N	O	F	Ne	Na	Mg	Al	Si	P	S	Cl	Ar	K	Ca
Mass No	1	4	7	9	11	12	14	16	19	20	23	24	27	28	31	32	35.5	40	39	40